

CSCI 618: Trustworthy ML (from an optimization lens)

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Fall 2026

Lecture 8, Sep 21

New training set



cat

cat

A simple Gaussian setting to understand tradeoff

$$y \sim^{u.a.r.} \{-1, +1\}, \quad x_1 = \begin{cases} +y, & \text{with probability } p, \\ -y, & \text{with probability } 1 - p, \end{cases} \quad x_2, \dots, x_{d+1} \stackrel{i.i.d.}{\sim} \mathcal{N}(\eta y, 1),$$

where $\mathcal{N}(\mu, \sigma^2)$ is a normal distribution with mean μ and variance σ^2 , and $p \geq 0.5$.

$$f_{\text{avg}}(x) := \text{sign}(w_{\text{unif}}^\top x), \quad \text{where } w_{\text{unif}} := \left[0, \frac{1}{d}, \dots, \frac{1}{d} \right],$$

Is f_{avg} robust?

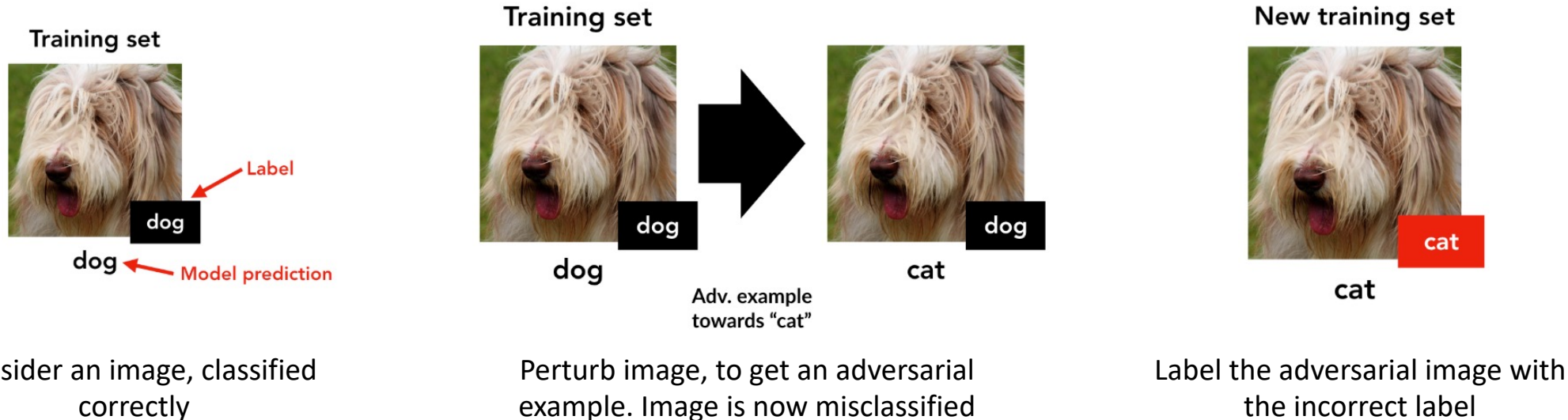
In this setting, we have

- Robust feature, x_1 : This has ℓ_∞ robustness even at $\epsilon = 0.99$, but only gets accuracy p
- Non-robust features $\{x_2, \dots, x_d\}$: Using these f_{avg} gets accuracy $>99\%$, but ℓ_∞ robustness only at $\epsilon \leq 2\eta$

Suppose $p = 0.95$. Then can show

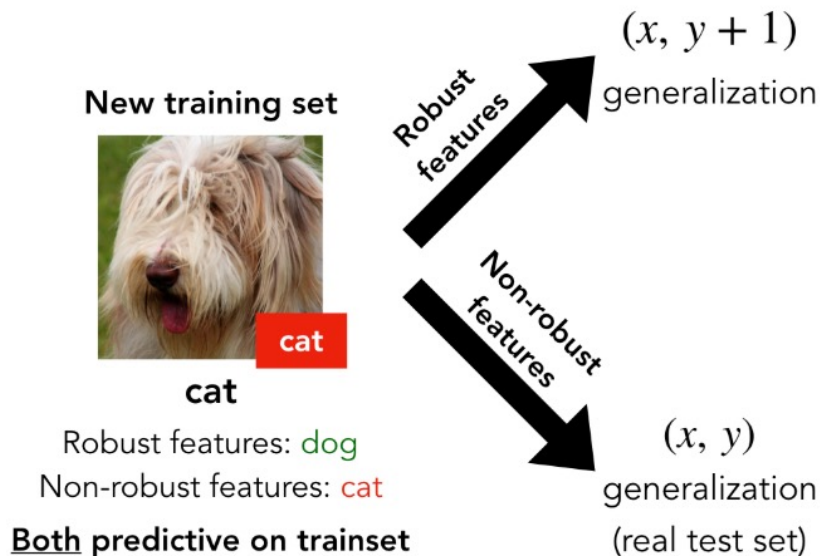
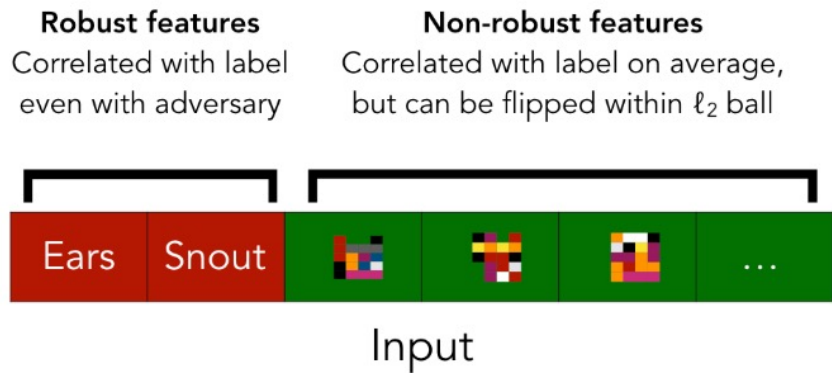
- If standard accuracy is much greater than 95%, say close to 100%, then robust accuracy is close to 0!
- Can get robust accuracy 95%, but only with standard accuracy at close to 95%!

A very cool experiment

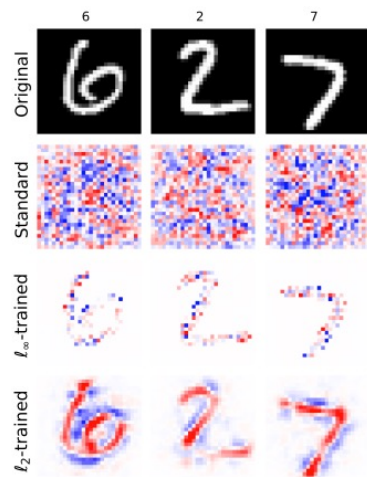


- Suppose we take CIFAR10 and a model trained on CIFAR10, replace each image by its adversarial example for some class, and “relabel” the image with this wrong class.
- Now train a model on this new CIFAR10, and then evaluate on the normal CIFAR10 test set. How much accuracy do we expect?
- Model gets highly non-trivial accuracy! ($\approx 45\%$ on 10 class classification)

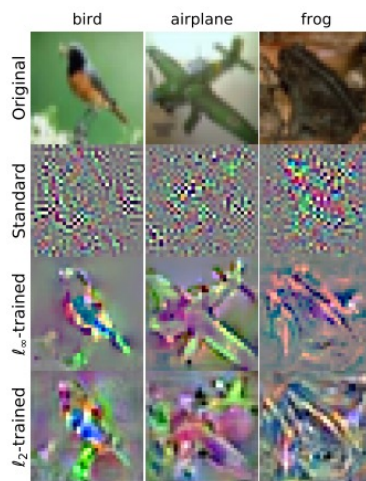
Explanation: Datasets have both robust and non-robust features



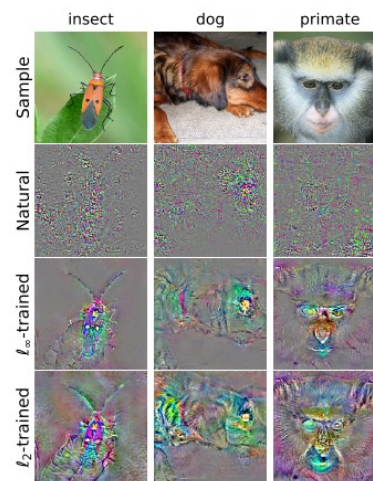
Robust training may learn representations more aligned with human perception



(a) MNIST



(b) CIFAR-10



(c) Restricted ImageNet

Visualization of the loss gradient with respect to input pixels. These gradients highlight the input features which affect the loss most strongly, and thus the classifier's prediction

Distributional robustness

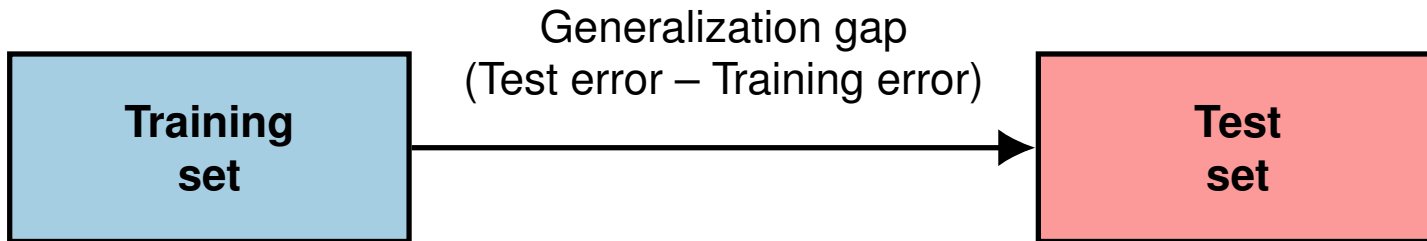


Distribution shifts: Setup

Recall that in supervised ML we care about expected loss (or the *risk*) under some distribution $\mathcal{D}$:

$$\begin{aligned} R(f) &= \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(f(x), y)] \\ &= \sum_{x', y'} \Pr_{\mathcal{D}}(x = x', y = y') \ell(f(x'), y'). \end{aligned}$$

We measure this with a test set.



What if we get training samples from $\mathcal{D}$, but test samples from $\mathcal{D}'$?

Distributionally robust optimization for subgroup robustness

In usual supervised ML we care about finding some predictor f^* such that

$$f^* := \arg \min_{f \in \mathcal{F}} \left\{ \mathbb{E}_{(x,y) \sim \mathcal{D}} [\ell(f(x), y)] \right\}.$$

Suppose we have a set of groups $g \in \mathcal{G}$, each of which defines some distribution $\mathcal{D}_g$ (which could be a re-weighting of $\mathcal{D}$ with respect to the marginal of x). Then we can define the distributionally robust formulation of ML as:

$$f_{\text{DRO}}^* := \arg \min_{f \in \mathcal{F}} \left\{ \max_{g \in \mathcal{G}} \mathbb{E}_{(x,y) \sim D_g} [\ell(f(x), y)] \right\}.$$

As is usual in supervised ML, we do not actually have access to the distribution D_g , but work with empirical samples.

$$\hat{f}_{\text{DRO}}^* := \arg \min_{f \in \mathcal{F}} \left\{ \max_{g \in \mathcal{G}} \frac{1}{|\text{\#samples from group } g|} \sum_{(x,y) \in \text{group } g} \ell(f(x), y) \right\}.$$

Also see *Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization*, Sagawa et al. '20

Algorithmic fairness

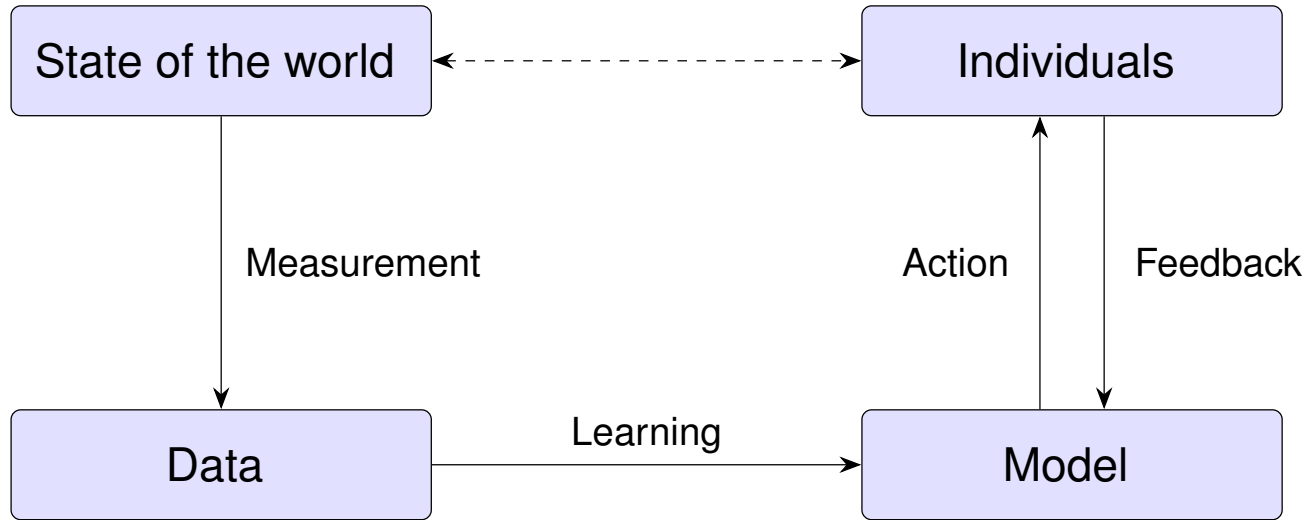
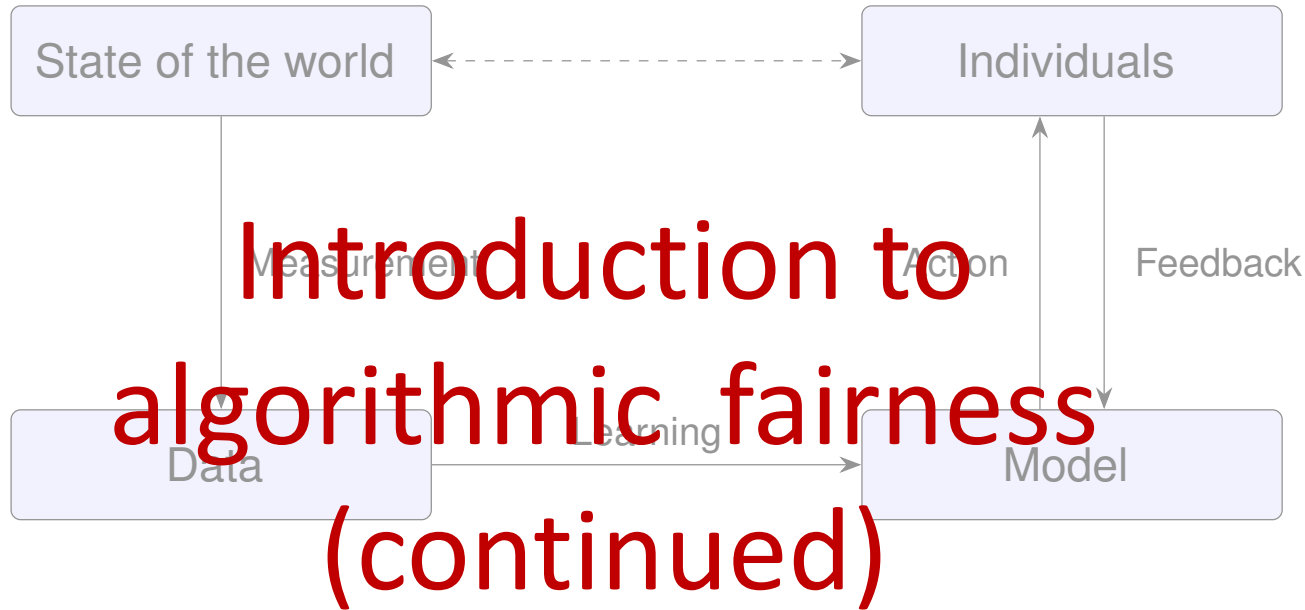


Fig. based on the book *Fairness And ML: Limitations and Opportunities*



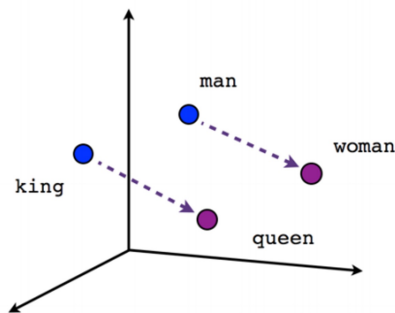
Unfairness could arise in various ways

- Unequal accuracy: The model may have poor performance on certain sub-populations or demographics
- Representation harm: The system may reinforce existing stereotype or biases
- Biased predictions: The predictions of the model could exhibit biases across different demographics
- ...

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Bias in representation: Word embeddings



Male-Female

Word analogy questions:

man: woman :: king : ??

$\xrightarrow{\text{woman}} - \xrightarrow{\text{man}} \approx \xrightarrow{\text{queen}} - \xrightarrow{\text{king}}$

$\xrightarrow{\text{woman}} - \xrightarrow{\text{man}} \approx \xrightarrow{\text{homemaker}} - \xrightarrow{\text{computer programmer}}$

Gender stereotype *she-he* analogies.

sewing-carpentry	register-nurse-physician	housewife-shopkeeper
nurse-surgeon	interior designer-architect	softball-baseball
blond-burly	feminism-conservatism	cosmetics-pharmaceuticals
giggle-chuckle	vocalist-guitarist	petite-lanky
sassy-snappy	diva-superstar	charming-affable
volleyball-football	cupcakes-pizzas	hairstylist-barber

Gender appropriate *she-he* analogies.

queen-king	sister-brother	mother-father
waitress-waiter	ovarian cancer-prostate cancer	convent-monastery

Bias in representation: Image generation

**a software
developer**



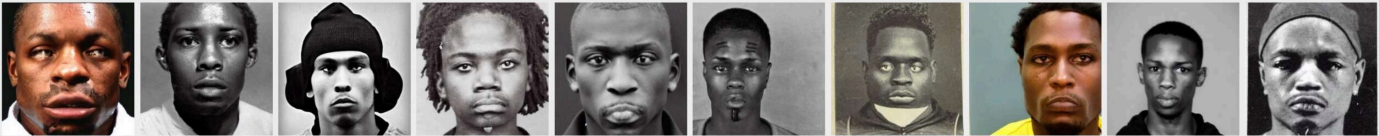
**a flight
attendant**



a terrorist



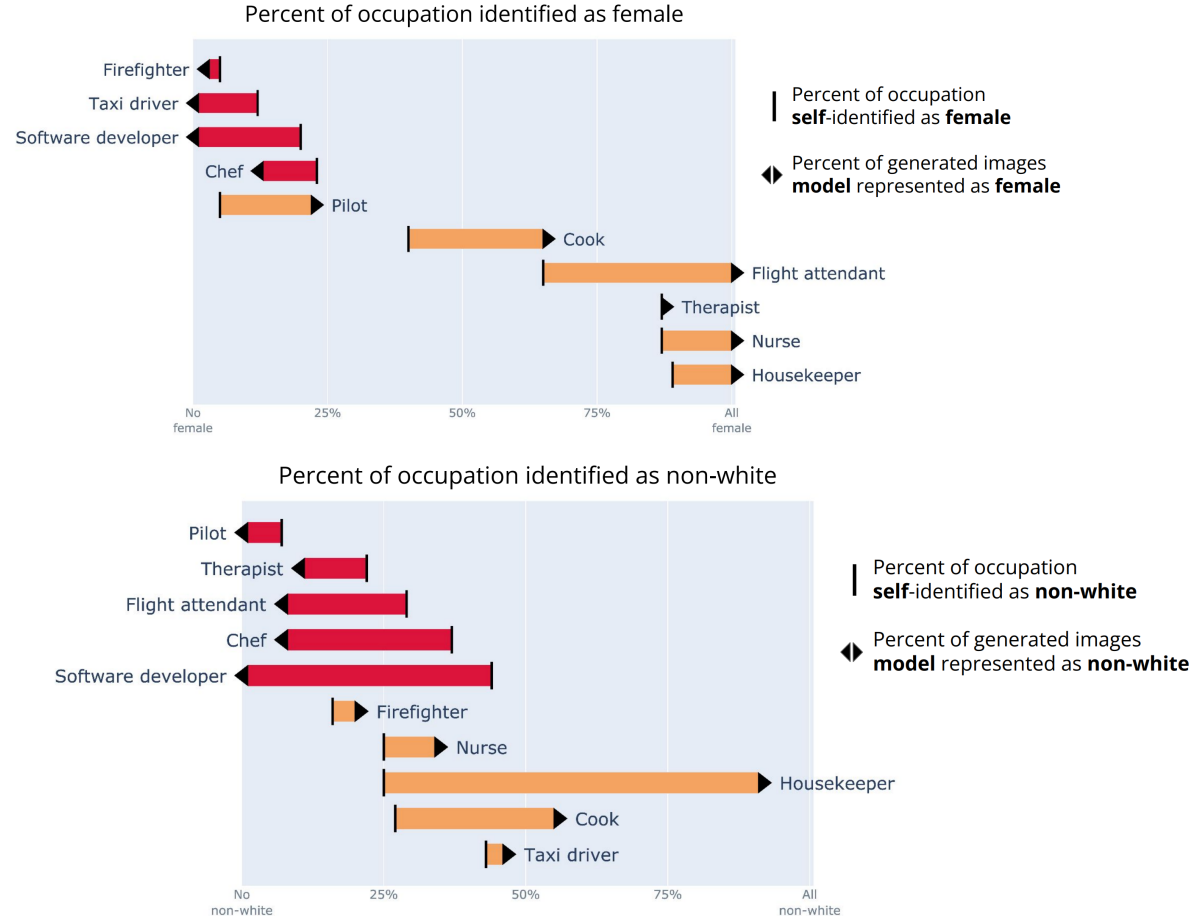
a thug



**an emotional
person**



Model amplifies existing biases



Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale, Bianchi et al., 2023
For more discussion, see *A Systematic Study of Bias Amplification*, Hall et al., 2022

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Bias in predictions: The COMPAS software

- COMPAS is a proprietary software used by many judicial systems to determine the risk that someone arrested for a crime again commits a crime in the future
- Used for decisions such as for deciding bail

Current Charges

- | | | | |
|---|--|---|---|
| ☐ Homicide | ☒ Weapons | ☒ Assault | ☐ Arson |
| ☐ Robbery | ☐ Burglary | ☐ Property/Larceny | ☐ Fraud |
| ☐ Drug Trafficking/Sales | ☐ Drug Possession/Use | ☐ DUI/DUIL | ☒ Other |
| ☐ Sex Offense with Force | ☐ Sex Offense w/o Force | | |

1. Do any current offenses involve family violence?

☒ No ☐ Yes

2. Which offense category represents the most serious current offense?

☐ Misdemeanor ☐ Non-violent Felony ☒ Violent Felony

3. Was this person on probation or parole at the time of the current offense?

☒ Probation ☐ Parole ☐ Both ☐ Neither

4. Based on the screener's observations, is this person a suspected or admitted gang member?

☐ No ☒ Yes

5. Number of pending charges or holds?

☒ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4+

6. Is the current top charge felony property or fraud?

☒ No ☐ Yes

Criminal History

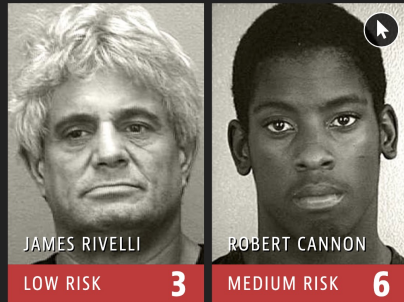
Exclude the current case for these questions.

Biases in COMPAS



<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

Two Shoplifting Arrests



After Rivelli stole from a CVS and was caught with heroin in his car, he was rated a low risk. He later shoplifted \$1,000 worth of tools from a Home Depot.

Two Drug Possession Arrests



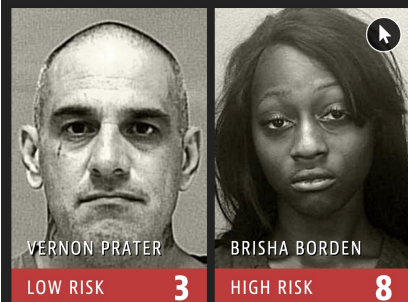
Fugett was rated low risk after being arrested with cocaine and marijuana. He was arrested three times on drug charges after that.

Two DUI Arrests



Lugo crashed his Lincoln Navigator into a Toyota Camry while drunk. He was rated as a low risk of reoffending despite the fact that it was at least his fourth DUI.

Two Petty Theft Arrests



Borden was rated high risk for future crime after she and a friend took a kid's bike and scooter that were sitting outside. She did not reoffend.

“In forecasting who would re-offend, the algorithm made mistakes with black and white defendants at roughly the same rate but in very different ways.

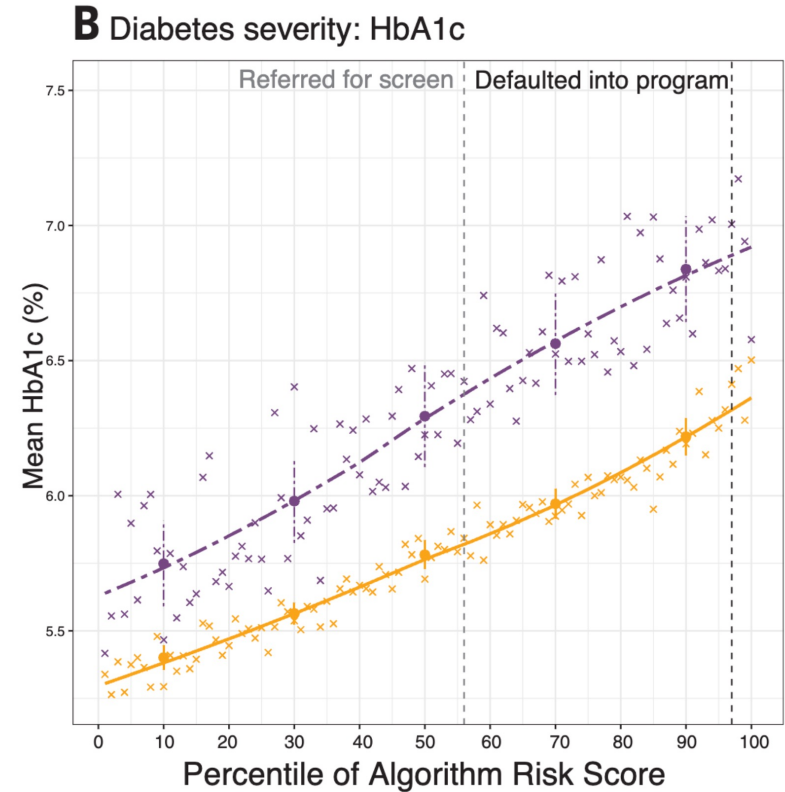
- The formula was particularly likely to falsely flag black defendants as future criminals, wrongly labeling them this way at almost twice the rate as white defendants.*
- White defendants were mislabeled as low risk more often than black defendants.”*

We will also see later that there are inherent tensions here: the COMPAS algorithm is biased in one way and unbiased in another, and it may be impossible to simultaneously be unbiased in both.

Bias in predictions: Predicting disease severity

Quoting from the paper:

- Health systems rely on commercial prediction algorithms to identify and help patients with complex health needs.
- A widely used algorithm affecting millions of patients, exhibits significant racial bias: At a given risk score, Black patients are considerably sicker than White patients, as evidenced by signs of uncontrolled illnesses.
- Remedying this disparity would increase the percentage of Black patients receiving additional help from 17.7 to 46.5%.
- Bias arises because the algorithm predicts health care costs rather than illness, but unequal access to care means typically less money was spent on care for Black patients than for White patients.



Dissecting racial bias in an algorithm used to manage the health of populations, Obermeyer et al., Science 2019

Some more instances of algorithmic decision making gone wrong

Aug 19, 2020 - Technology

How an AI grading system ignited a national controversy in the U.K.



Bryan Walsh, author of [Axios Future](#)



Illustration: Eniola Odetunde/Axios

A huge controversy in the U.K. over an algorithm used to substitute for university-entrance exams highlights problems with the use of AI in the real world.

[Link to article](#)



[Link to article](#)

Some more instances of algorithmic decision making gone wrong

The New York Times

There Is a Racial Divide in Speech-Recognition Systems, Researchers Say

Technology from Amazon, Apple, Google, IBM and Microsoft misidentified 35 percent of words from people who were black. White people fared much better.



Amazon's Echo device is one of many similar gadgets on the market. Researchers say there is a racial divide in the usefulness of speech recognition systems. Grant Hinsley for The New York Times

[Link to article](#)

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ARTIFICIAL INTELLIGENCE

LinkedIn's job-matching AI was biased. The company's solution? More AI.

ZipRecruiter, CareerBuilder, LinkedIn — most of the world's biggest job search sites use AI to match people with job openings. But the algorithms don't always play fair.

By Sheridan Wall & Hilke Schellmann

June 23, 2021



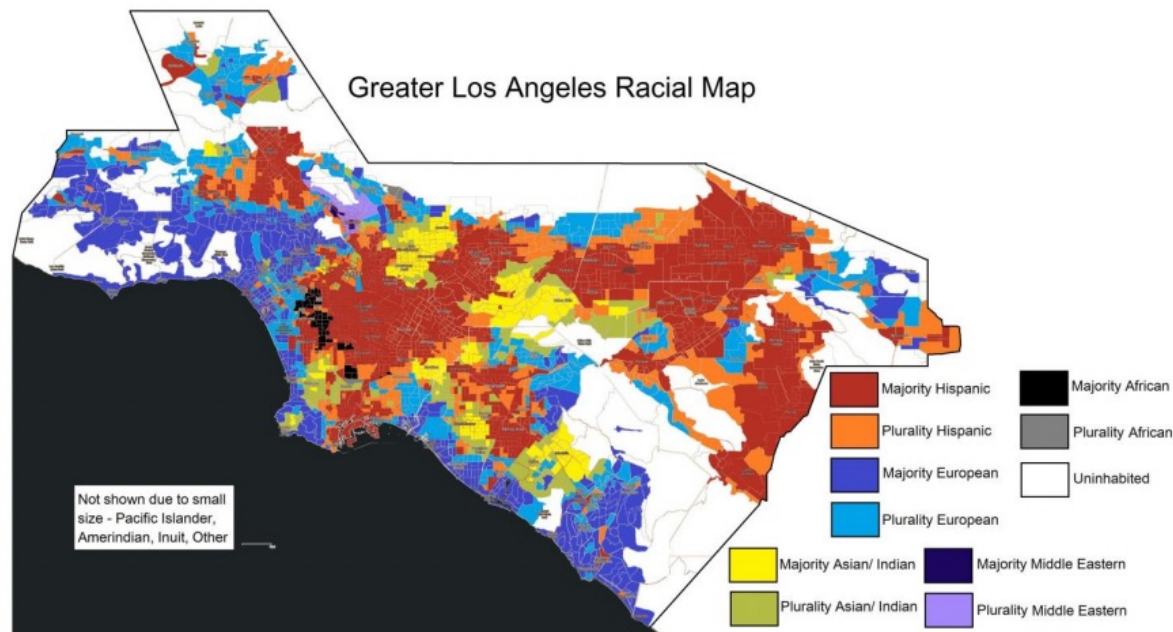
[Link to article](#)



How to obtain fair classifiers?

Observation: No fairness by just excluding sensitive attributes

Why? Sensitive attribute can often be reconstructed from other features



Zip code has a lot of information about race

Ensuring fairness in classification: **Group & Individual fairness notions**

Two broad classes of fairness notions in classification:

Individual fairness: Algorithm treats **similar individuals similarly**

Group fairness: Algorithm is **“unbiased” on protected groups** (such as race, gender etc.)

Individual fairness

Define a **metric** $d(x, x')$ for the similarity between any two individuals x and x' .

e.g.: $d(x, x') = \|x - x'\|_2$

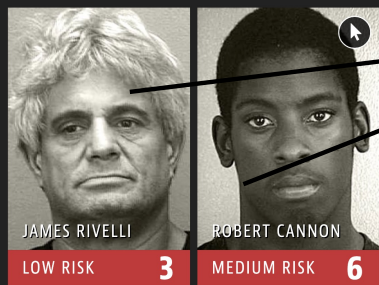
If classifier predicts $p(x)$ as the probability of label being one for x , if

$$|p(x) - p(x')| \leq \mu d(x, x'),$$

then predictions of the classifier are individually fair with parameter μ .



Two Shoplifting Arrests



After Rivelli stole from a CVS and was caught with heroin in his car, he was rated a low risk. He later shoplifted \$1,000 worth of tools from a Home Depot.

If these two individuals are similar, then their risk scores should be similar.

Group fairness

Group fairness notions require that the models predictions obey certain properties over protected groups (e.g. by race, gender).

Many different notions have been proposed

- Statistical parity
- Equalized odds
- Calibration across groups

Statistical parity/Demographic parity

Binary classification setup (e.g. admitting a student to a degree program)

- Classifier f
- Datapoint (x, y)
- Sensitive attribute $a \in \{0, 1\}$

Statistical parity (also known as demographic parity): $\Pr_x[f(x) = 1 \mid a = 1] = \Pr_x[f(x) = 1 \mid a = 0]$

In words: **Predictions are independent of sensitive attribute**

E.g., admit equal fraction of men or women into program

Can be too strong if labels and sensitive attribute are not independent.

E.g. if one demographic is more likely to be qualified than the other

Equalized odds & Equality of opportunity

Same binary classification setup (e.g. admitting student to degree program)

- Classifier f
- Datapoint (x, y)
- Sensitive attribute $a \in \{0, 1\}$

Equalized odds: Following 2 constraints are satisfied

1. $\Pr_{x,y}[f(x) = 1 \mid a = 1, y = 1] = \Pr_{x,y}[f(x) = 1 \mid a = 0, y = 1]$
2. $\Pr_{x,y}[f(x) = 0 \mid a = 1, y = 0] = \Pr_{x,y}[f(x) = 0 \mid a = 0, y = 0]$

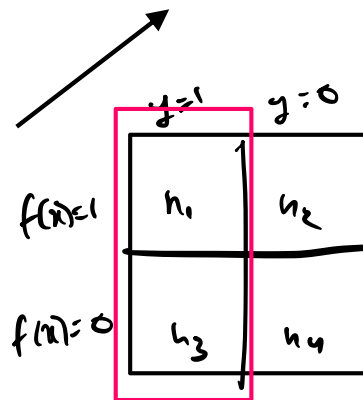
Equivalently:

Recall for $y = 1$ is the same for both groups (1st condition)

False positive rate (**FPR**) is the same for both groups (2nd condition)

Also equivalent to saying: Conditioned on label, prediction is independent of sensitive attribute

Just having 1st constraint (for $y = 1$) gives **equality of opportunity**



Recall for class 1 =
 $\Pr_{x,y}[f(x) = 1 \mid y = 1]$

FPR =
 $\Pr_{x,y}[f(x) = 1 \mid y = 0]$

Equalized odds

E.g. Professor Snape has to admit students to his Advanced Potions class.

100 students apply from Slytherin (80% are qualified)

	Qualified	Unqualified
Accepted	60	5
Rejected	20	15
Total	80	20

100 students apply from Gryffindor (40% are qualified)

	Qualified	Unqualified
Accepted	30	15
Rejected	10	45
Total	40	60

Is Prof. Snape fair based on

- (i) statistical parity,
- (ii) equalized odds?

Statistical parity :

$$\Pr_{x,y}[f(x) = 1 \mid a = 1] = \Pr_{x,y}[f(x) = 1 \mid a = 0]$$

Equalized odds:

$$\Pr_{x,y}[f(x) = 1 \mid a = 1, y = 1] = \Pr_{x,y}[f(x) = 1 \mid a = 0, y = 1]$$

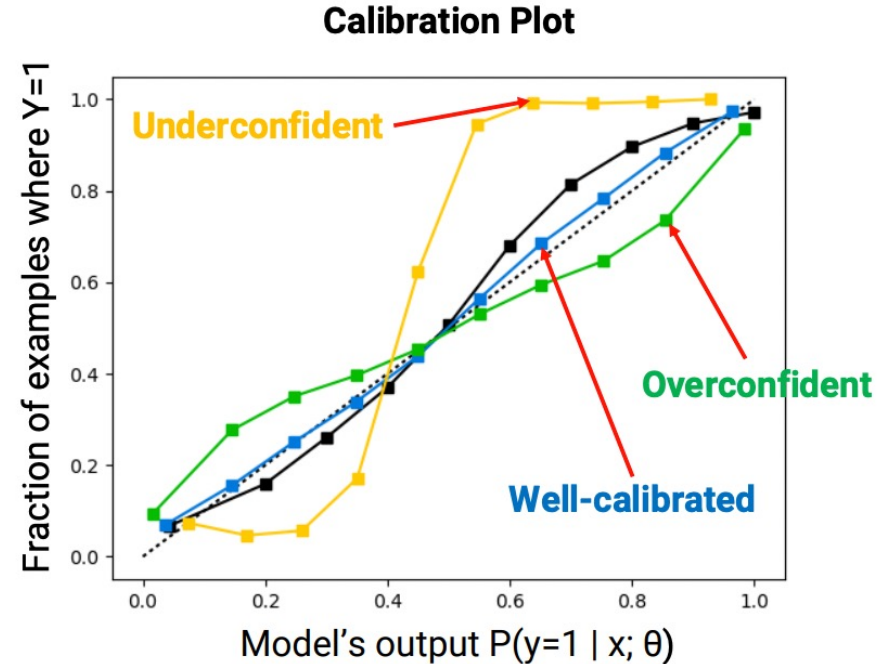
$$\Pr_{x,y}[f(x) = 0 \mid a = 1, y = 0] = \Pr_{x,y}[f(x) = 0 \mid a = 0, y = 0]$$

Calibration

Calibration: A model f for binary classification is calibrated if

$$\Pr_{x,y}[y = 1 \mid f(x) = \alpha] = \alpha$$

Informally, this says that “predictions mean what they should”



This is known as a **reliability diagram**

Calibration across groups

A model f for binary classification is calibrated for groups defined by sensitive attribute a if

$$\Pr_{x,y}[y = 1 \mid f(x) = \alpha, a = 1] = \alpha,$$

$$\Pr_{x,y}[y = 1 \mid f(x) = \alpha, a = 0] = \alpha.$$

Informally, this says that “predictions mean what they should **for each group**”

